

WORK SHEET - 06

P-BLOCK ELEMENTS

Single correct :

- Q.1 Match the following mixtures with the respective solution used for their separation:
- | | | |
|------------------------------------|----------------------|---------------------------------------|
| (a) N_2 & CO | (i) water | |
| (b) N_2 & O_2 | (ii) H_2SO_4 | |
| (c) N_2 & NH_3 | (iii) Ammonical CuCl | |
| (d) PH_3 & NH_3 | (iv) Pyrogallol | |
| (A) a-(iv), b-(iii), c-(ii), d-(i) | | (B) a-(iii), b-(iv), c-(ii)(i), d-(i) |
| (C) a-(iii), b-(iv), c-(i), d-(ii) | | (D) a-(iii), b-(ii), c-(iv), d-(i) |
- Q.2 Which option is **correct** for hydrolysis of PCl_5 ?
- | | | |
|--|----------------------|---------------------------|
| (a) Produce H_3PO_4 | (b) Produce $POCl_3$ | (c) Produce Tribasic acid |
| (d) 4 water molecules are utilized for complete hydrolysis | | |
| (A) TTTT | (B) TTFT | (C) FTTF |
| | | (D) TTFF |
- Q.3 Select the **correct** statement regarding $SnCl_2$ and $SnCl_4$.
- (A) $SnCl_2$ is more covalent than $SnCl_4$
 (B) $SnCl_2$ is more volatile than $SnCl_4$
 (C) $SnCl_2$ is more soluble in anhydrous ethanol than $SnCl_4$
 (D) Rate of hydrolysis of $SnCl_4$ is higher than that of $SnCl_2$
- Q.4 $B_3N_3H_6$ upon hydrolysis yields
- | | | | |
|---------------|------------|-----------|------------------|
| (A) H_3BO_3 | (B) NH_3 | (C) H_2 | (D) All of these |
|---------------|------------|-----------|------------------|
- Q.5 Consider following transformations :
- (I) $XeF_6 + NaF \longrightarrow Na^+[XeF_7]^-$
 (II) $Al_2O_3(s) + 6HCl(aq) + 9H_2O(l) \longrightarrow 2[Al(H_2O)_6]^{3+}(aq) + 6Cl^-(aq)$
 (III) $Al_2O_3(s) + 6NaOH(aq) + 3H_2O(l) \longrightarrow 2Na_3[Al(OH)_6](aq)$
 Possible transformations are :
- | | | | |
|----------------|------------|-----------|-------------|
| (A) I, II, III | (B) I, III | (C) I, II | (D) II, III |
|----------------|------------|-----------|-------------|
- Q.6 Which of the following does not exist?
- | | | | |
|---------------|-------------|--------------|--------------|
| (A) $Tl[I_3]$ | (B) PBr_5 | (C) $SbCl_5$ | (D) $BiBr_5$ |
|---------------|-------------|--------------|--------------|
- Q.7 Select the correct statements regarding the products obtained when $Na_2S_2O_3 \cdot 5H_2O$ is subjected to strong heating.
- (I) Disproportionation reaction takes place
 (II) Na_2SO_3 and Na_2S_5 are produced.
 (III) Na_2SO_4 and Na_2S_5 are produced.
 (IV) One of the products has zig-zag open chain structure.
- | | | | |
|---------------|----------------|-----------------|--------------------|
| (A) I, II, IV | (B) I, III, IV | (C) II, III, IV | (D) I, II, III, IV |
|---------------|----------------|-----------------|--------------------|
- Q.8 The substrate which has maximum rate of hydrolysis, is:
- | | | | |
|-------------|--------------|------------|-------------|
| (A) CCl_4 | (B) $SiCl_4$ | (C) SF_6 | (D) PCl_5 |
|-------------|--------------|------------|-------------|
- Q.9 The product formed in the reaction of $SOCl_2$ with white phosphorous is
- | | | | |
|-------------|----------------|-------------|--------------|
| (A) PCl_3 | (B) SO_2Cl_2 | (C) SCl_2 | (D) $POCl_3$ |
|-------------|----------------|-------------|--------------|
- Q.10 The interhalogen compound that cannot exist is
- | | | | |
|-----------|-------------|------------|-------------|
| (A) IBr | (B) ICl_3 | (C) IF_4 | (D) BrF_5 |
|-----------|-------------|------------|-------------|
- Q.11 The reagent which results in symmetrical cleavage of diborane is :
- | | | | |
|---------------------|-----------------|-------------------|-------------|
| (A) ethylenediamine | (B) methylamine | (C) dimethylamine | (D) ammonia |
|---------------------|-----------------|-------------------|-------------|

- Q.12 Select the reaction which describes the existence of $I_2O_5(s)$ as $(IO_2^+)(IO_3^-)$
 (A) $I_2O_5 + HF \longrightarrow HIO_2 + FIO_3$ (B) $I_2O_5 + HF \longrightarrow IO_2F + HIO_3$
 (C) $I_2O_5 + HF \longrightarrow HOI + IO_4F$ (D) $I_2O_5 + HF \longrightarrow IOF + HIO_4$
- Q.13 Which of the following oxy-acids of nitrogen exists in dimeric form?
 (A) Nitrous acid (B) Hypo nitrous acid (C) Per nitric acid (D) Nitric acid
- Q.14 $P_2O_5 + H_2O \longrightarrow X$
 $2X - H_2O \longrightarrow Y$
 $Y + O \longrightarrow Z$
 $X + O \longrightarrow E$
 $X - O \longrightarrow F$
 Find the sum of π bond in X, Y, Z, E and F ?
 (A) 6 (B) 7 (C) 5 (D) 8
- Q.15 $NH_3 + Cl_2(\text{excess}) \rightarrow \text{Explosive substance (X)} \xrightarrow{HCl(aq)} (Y) + (Z)$
 $(Y) \xrightarrow{NiCl_2} \text{Blue}$
 The products (Y) and (Z) are respectively:
 (A) NH_3 & Cl_2 (B) N_2 & Cl_2 (C) $NH_2 - NH_2$ & Cl_2 (D) $HN = NH$ & Cl_2
- Q.16 Which of the following will give H_2 gas with dilute HNO_3 ?
 (A) Mg (B) Zn (C) Cu (D) Hg
- Q.17 $\begin{array}{c} \text{COOH} \\ \diagup \\ CH_2 \\ \diagdown \\ \text{COOH} \end{array} \xrightarrow[P_4O_{10}]{\Delta} 'A' + H_2O$
 How many number of 'sp' hybridised atoms present in 'A'?
 (A) 3 (B) 5 (C) 2 (D) None of these
- Q.18 The compound which undergoes disproportionation in alkaline solution is :
 (A) N_2O_3 (B) Cl_2O_6 (C) N_2O_5 (D) Cl_2O_7
- Q.19 Consider the following compounds.
 (I) XeF_2 (II) XeF_4 (III) XeF_6
 The **correct** order of tendency to accept F^- ion is :
 (A) $I > II > III$ (B) $II > I > III$ (C) $I > III > II$ (D) $III > II > I$
- Q.20 Violet colour of iodine is decolourised by :
 (A) Hot and Concentrated NaOH (B) $Na_2S_2O_3$
 (C) Concentrated HNO_3 (D) All of these
- Q.21 Under ambient conditions, the total number of gases released as products in the final step of the reaction scheme shown below is
- $$XeF_6 \xrightarrow{\text{Complete Hydrolysis}} P + \text{other product}$$

$$\downarrow OH^- / H_2O$$

$$Q$$

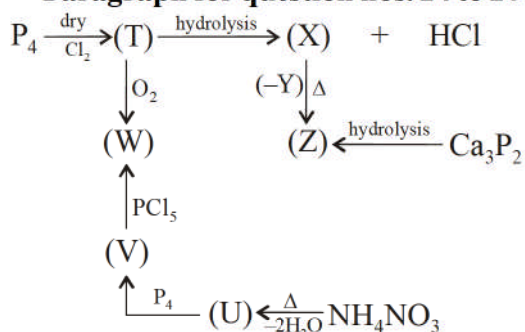
$$\downarrow \text{slow disproportionation in } OH^- / H_2O$$

$$\text{Products}$$
- (A) 0 (B) 1 (C) 2 (D) 3

- Q.22 The reaction of P_4 with X leads selectively to P_4O_6 . The X is -
 (A) Dry O_2 (B) A mixture of O_2 and N_2
 (C) Moist O_2 (D) O_2 in the presence of aqueous NaOH
- Q.23 When PH_3 absorbed in $HgCl_2$ solution the corresponding phosphide is obtained
 (A) Hg_2P_3 (B) Hg_3P_2 (C) $Hg_3(PO_4)_2$ (D) None of these

Comprehension

Paragraph for question nos. 24 to 26



- Q.24 The compound (W) can also be obtained by the reaction(s).
 (A) $PCl_5 + H_2SO_4 \longrightarrow ?$ (B) $PCl_5 + SO_2 \longrightarrow ?$
 (C) $PCl_5 + H_2O \longrightarrow ?$ (D) $PCl_5 + O_2 \longrightarrow ?$
 (Equimolar ratio)
- Q.25 Select the correct statement(s)?
 (A) Heating of (X) involves disproportionation.
 (B) Heating of NH_4NO_3 to get (U) involves disproportionation
 (C) The compound (U) supports combustion.
 (D) Formation of (Z) by treating P_4 with caustic soda involves disproportionation.
- Q.26 The gas (U) is
 (A) neutral oxide
 (B) oxidising agent
 (C) also obtained by treating zinc with dilute HNO_3
 (D) linear in shape.

Paragraph for question nos. 27 to 29

	F	Cl	Br
B	Monomer	Monomer	X
Al	Ionic	Polymer (W)	Y
Ga	Ionic	Dimer	Z

- Q.27 The compound (X) exists as
 (A) Monomer (B) Dimer (C) Polymer (D) Trimer
- Q.28 Select the correct statement(s) regarding the compounds (W) and (Z).
 (A) Coordination number of Al in the compound (W) is 6
 (B) Coordination number of Ga in the compound (Z) is 4.
 (C) The compound (Z) exists as dimer.
 (D) All of these
- Q.29 Which of the following statements is correct for the compound (Y)?
 (A) It exists as polymer and the coordination number of Al is 6.
 (B) It exists as dimer and the coordination number of Al is 4.
 (C) It exists as monomer.
 (D) None of these

Assertion and Reason :

Q.30 **Statement-I :** $\text{NaBO}_3/\text{OH}^-$ can be used for oxidation of Cr^{3+} to Cr^{6+}

Statement-II : In alkaline medium NaBO_3 produces H_2O_2

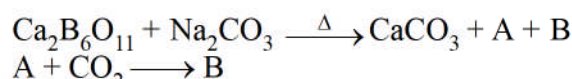
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

More than one may be correct

Q.31 Which of the following option is/are incorrect for the most acidic halide among Boron halides?

- (A) It is volatile halide. (B) It is most thermal stable halide.
(C) It form strong $p\pi-p\pi$ back bond. (D) It can form $[\text{BX}_4^-]$

Q.32 Which of the following option is / are correct?



- (A) B contains 5 B–O–B linkage
(B) A will be obtained by thermal decomposition of B
(C) Aqueous solution of B used as a acidic buffer in titration
(D) A exist in a polymeric form.

Q.33 Which of the following undergo disproportionation on heating?

- (A) H_3PO_3 (B) Na_3PO_4 (C) H_2SO_3 (D) H_3BO_3

Q.34 Consider the following reactions.

- (I) $\text{PH}_3 + \text{HI} \longrightarrow \text{PH}_4^+ \text{I}^-$
(II) $\text{NH}_3 + \text{HI} \longrightarrow \text{NH}_4^+ \text{I}^-$
(III) $\text{BF}_3 + \text{LiF} \longrightarrow \text{Li}^+ \text{BF}_4^-$

Select the correct options:

- (A) In I, $\hat{\text{H}}\text{P}\text{H}$ in PH_3 is less than that in PH_4^+
(B) In II, $\hat{\text{H}}\text{N}\text{H}$ in NH_3 less than that in NH_4^+
(C) Reaction (I) involves maximum increase in bond angle.
(D) Reactions (I), II and (III) involve increase in bond angle.

Q.35 Select the correct statement(s) regarding O_3 and O_2 .

- (A) O_3 is diamagnetic and O_2 is paramagnetic
(B) O_3 is more soluble in water than O_2
(C) O_3 is polar and O_2 is non-polar molecule.
(D) O_3 has higher boiling point than O_2

Q.36 Which of the following options is / are correct?

- (A) $\text{POF}_3 > \text{POCl}_3 > \text{POBr}_3 > \text{POI}_3$ (P–O Bond length)
(B) $\text{H}_2\text{O} < \text{CH}_3\text{OH} < \text{CH}_3\text{OCH}_3$ (Bond angle)
(C) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ (Oxidising property)
(D) $\text{Xe} > \text{Kr} > \text{Ar} > \text{Ne} > \text{He}$ (Solubility in water)

Q.37 Choose the reactions which would liberate nitrogen gas?

- (A) $\text{Ca}(\text{OCl})\text{Cl} + \text{NH}_3 \xrightarrow[\text{medium}]{\text{aqueous}} ?$ (B) $\text{NH}_3 + \text{PbO} \xrightarrow{\Delta} ?$
(C) $\text{NH}_3 (\text{excess}) + \text{Cl}_2 \xrightarrow{\Delta} ?$ (D) $\text{NH}_4\text{Cl} + \text{NaNO}_2 \xrightarrow{\Delta} ?$

Q.38 The compounds which do not undergo hydrolysis at room temperature are

- (A) NF_3 (B) CCl_4 (C) SiH_4 (D) SF_6



Match the column

Q.39	Column I (Hydrolysis of substrates)	Column II (Products of hydrolysis & nature of hydrolysis)
(A)	ICl_3	(P) HIO_4
(B)	I_2O_7	(Q) HIO_2
(C)	IF_5	(R) HIO_3
(D)	I_2O_6	(S) Disproportionation
		(T) Non - redox

Q.40	Column-I (Reactions)	Column-II (Products and type of reaction)
(A)	$\text{P}_4\text{O}_6 + \text{H}_2\text{O} \longrightarrow$	(P) H_3PO_3
(B)	$\text{P}_4 + \text{NaOH (aq.)} \xrightarrow{\Delta}$	(Q) H_3PO_4
(C)	$\text{H}_3\text{PO}_3 \xrightarrow{\Delta}$	(R) PH_3
		(S) disproportionation.

Q.41	Match the column-	
	Column-I (Reactions)	Column-II (Product)
(A)	Borax $\xrightarrow{\Delta}$	(p) BN
(B)	$\text{B}_2\text{H}_6 + \text{H}_2\text{O} \longrightarrow$	(q) B_2H_6
(C)	$\text{B}_2\text{H}_6 + \text{NH}_3$ (Excess) $\xrightarrow{\Delta}$	(r) H_3BO_3
(D)	$\text{BCl}_3 + \text{LiAlH}_4 \longrightarrow$	(s) $\text{NaBO}_2 + \text{B}_2\text{O}_3$

Q.42	Match each of reactions given in Column I with the corresponding product(s) given in Column II
Column I	Column II
(A) $\text{Cu} + \text{dil HNO}_3$	(p) NO
(B) $\text{Cu} + \text{conc HNO}_3$	(q) NO_2
(C) $\text{Zn} + \text{dil HNO}_3$	(r) N_2O
(D) $\text{Zn} + \text{conc HNO}_3$	(s) $\text{Cu}(\text{NO}_3)_2$
	(t) $\text{Zn}(\text{NO}_3)_2$

Subjective

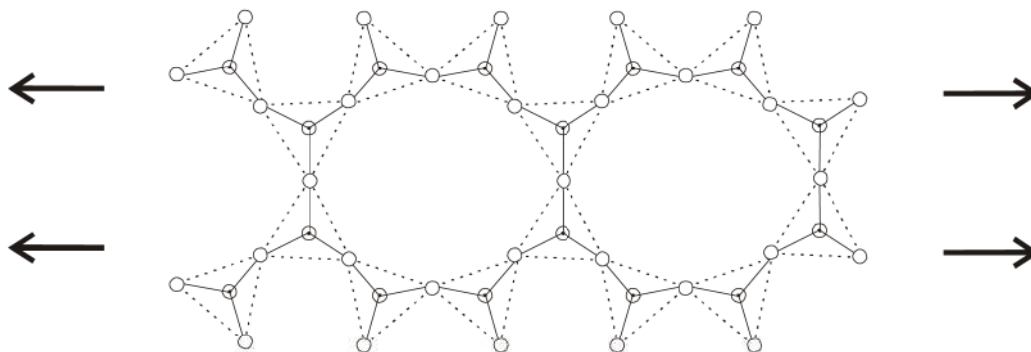
Q.43 How many of the following reactions would produce phosphine gas?

- | | |
|--|--|
| (a) $\text{PH}_4\text{Cl} + \text{NaOH} \longrightarrow$ | (b) $\text{P}_4 + \text{NaOH} \xrightarrow{\Delta}$ |
| (c) $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \longrightarrow$ | (d) $\text{H}_3\text{PO}_3 \xrightarrow{\Delta}$ |
| (e) $\text{PCl}_5 + \text{H}_2\text{SO}_4 \longrightarrow$ | (f) $\text{H}_3\text{PO}_2 \xrightarrow{\Delta}$ |
| (g) $\text{P}_2\text{Cl}_4 + \text{H}_2\text{O} \longrightarrow$ | (h) $\text{AlP} + \text{H}_2\text{SO}_4 \longrightarrow$ |
| (i) $\text{P}_4\text{O}_{10} + \text{H}_2\text{O} \longrightarrow$ | |

If your answer is 3, write the answer as 0003.

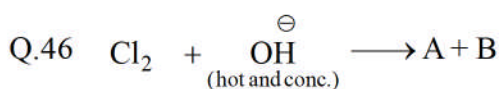
Q.44 When the compound (X) is treated with sodium iodate and hydrochloric acid, a violet coloured substance (Z) is produced. The violet colour of (Z) disappears when treated with the compound (A), producing (X) and (X'). (A) is produced by reaction between sulphur and sodium sulphite. Find the maximum number of bonds between identical atoms having equal bond length in the compound (X').

Q.45



In above unknown silicate structure, if general formula of Silicate is Si_mO_n^q .

If the value of $m = 6$ then identified the value of $(n + q)$ is -



Find the sum of oxidation state of Cl in A and B.

Q.47 Find the number of compounds which liberate NO_2 gas as one of the products on heating.
 NH_4NO_3 , NH_4NO_2 , $\text{Ca}(\text{NO}_3)_2$, $\text{Zn}(\text{NO}_3)_2$, $\text{Hg}(\text{NO}_3)_2$, AgNO_3 , LiNO_3 , $\text{Al}(\text{NO}_3)_3$, KNO_3
If your answer is 4, write the answer as 0004.

Q.48 Find the number of reaction, which will give ammonia.



ANSWER KEY

Q.1	B	Q.2	A	Q.3	D	Q.4	D	Q.5	A
Q.6	D	Q.7	B	Q.8	D	Q.9	A	Q.10	C
Q.11	A	Q.12	B	Q.13	B	Q.14	B	Q.15	A
Q.16	A	Q.17	A	Q.18	B	Q.19	D	Q.20	D
Q.21	C	Q.22	B	Q.23	B	Q.24	ABC	Q.25	ACD
Q.26	ABCD	Q.27	A	Q.28	D	Q.29	B	Q.30	A
Q.31	BCD	Q.32	ABD	Q.33	AC	Q.34	ABC		
Q.35	ABCD	Q.36	BCD	Q.37	ABCD	Q.38	ABD		
Q.39	(A) QT (B) PT (C) RT (D) PRS	Q.40	(A) P, (B) RS, (C) QRS						
Q.41	(A) s (B) r (C) p (D) q	Q.42	(A) p, s ; (B) q, s (C) r, t ; (D) q, t	Q.43	0006				
Q.44	2	Q.45	7	Q.46	0004	Q.47	0006	Q.48	7

